

Example 13.7

For the A.P given by 5, 3.5, 2, 0.5, -1, -2.5, ...

- i) Find the sum of the first 10 terms
- ii) Find the sum of the terms from 11th to 20th, both terms included

Solution:

- i) Using $a=5$, $d=-1.5$ and $n=10$ in (I), we get

$$S_{10} = (10/2) [2(5) + (10-1)(-1.5)] = 5[10-9(1.5)] = -17.5$$

- ii) The required sum $= S_{20} - S_{10}$

Using $a=5$, $d=-1.5$ and $n=20$ in (I), we get

$$S_{20} = (20/2)[2(5) + (20-1)(-1.5)] = 10[10-19(1.5)] = -185$$

$$\text{Thus, the required sum} = S_{20} - S_{10} = -185 - (-17.5) = -167.5$$

B) Geometric Progression (G.P)

For some constant real numbers a and r , the sequence

$$a, ar, ar^2, ar^3, \dots$$

is called a **Geometric Sequence or Geometric Progression (G.P)**.

Then the number r is called the **common ratio** and the number a is called the **scale factor**.

We may assume that $a \neq 0$ and $r \neq 0$, and $r \neq 1$ because

- i) $a=0$, gives the trivial sequence 0, 0, 0, ...,
- ii) $r=0$ gives the trivial sequence $a, 0, 0, \dots$, and also,
- iii) $r=1$ gives the trivial sequence $a, a, a, \dots$

n^{th} term of G.P: It can be easily verified that **n^{th} term t_n** of G. P., viz., **(t_n)** —with **initial term as $t_1=a$** , and **common ratio as r** —is given by **$t_n = ar^{(n-1)}$**

Example 13.8

- i) Sequence 3, 6, 12, 24, ... is G.P with initial term as 3, and common ratio as 2,
- ii) Sequence 3, -6, 12, -24, ... is G.P with initial term as 3, and common ratio as (-2)
- iii) Sequence 6, 2, 2/3, 2/9, 2/27, ... is a G. P. with initial term as 6, and common ratio as (1/3)

$$40^{\text{th}} \text{ term of sequence (i)} = (3)(2)^{(40-1)} = 3 \times 2^{39}$$

$$40^{\text{th}} \text{ term of sequence (ii)} = (3)(-2)^{(40-1)} = -3 \times 2^{39}$$

$$\text{However, } 39^{\text{th}} \text{ term of sequence (i)} = 39^{\text{th}} \text{ term of sequence (ii)} = 3 \times 2^{38}$$

$$21^{\text{st}} \text{ term of sequence (iii)} = (6) \times (1/3)^{(21-1)} = 6 \times (1/3)^{20} = 2 \times (1/3)^{19}$$

Sum S_n of first n terms of a G.P

$$(t_1, t_2, \dots, t_n, \dots) = (a, a \times r, \dots, a \times r^{n-1}, \dots)$$

is given by

$$S_n = a [1 + r + \dots + r^{n-1}]$$

$$= a (r^n - 1) / (r - 1) \dots \dots (I)$$

When $|r| < 1$, we may rewrite (II) as

$$S_n = a (1 - r^n) / (1 - r) \dots \dots (II)$$

Example 13.9

For the sequence given by 3, -6, 12, -24, ...

- Find the sum of the first 10 terms
- Find the sum of the first 11 terms
- Find the sum of the terms from 11th to 20th, both terms included

Solution:

The given sequence is a G. P with initial term $a = 3$, and common ratio as (-2) , and

for (i) $n = 10$. Using (II), we get

$$S_{10} = 3 \times [(-2)^{10} - 1] / (-2 - 1) = -[2^{10} - 1]$$

for (ii) $n = 11$. Using (II), we get

$$S_{11} = 3 \times [(-2)^{11} - 1] / (-2 - 1) = (2^{11} + 1).$$

For (iii), we need to calculate

$$S_{20} - \{S_{10}\} = \{3 [(-2)^{20} - 1] / (-2 - 1)\} - \{-[2^{10} - 1]\} = -[2^{20} - 1] + [2^{10} - 1]$$

$$= -2^{20} + 2^{10} = 2^{10} \{-2^{10} + 1\} = 2^{10}(1 - 2^{10}).$$

C) Harmonic Progression (H.P)

A Harmonic Progression, or harmonic sequence, is a sequence obtained by taking **the reciprocals of an arithmetic progression**, say, $a, a + d, a + 2d, a + 3d, \dots$

with some constant numbers a and d such that *none* of $a + n \times d = 0$, for $n = 0, 1, 2, \dots$,

because, H.P consists of terms as $1/(a + n \times d)$.

Some of the facts about an H.P may be derived from the corresponding facts about A.P

n^{th} term of an H. P.: As n^{th} term of A. P.—with *initial term as a* , and *common difference as d* —is given by $a + (n-1) \times d$,

the n^{th} term of an H. P. = $1 / (a + (n-1) \times d)$

Example 13.10

The examples are through corresponding A.P's

- i) For the A. P. 1, 2, 3, 4, ... (for $a=1$, and $d=1$),

The corresponding H.P

$$1, 1/2, 1/3, \dots, 1/n, \dots$$

is also known as the *Harmonic Sequence*

- ii) For the A. P. with $a = 1/16$, and $d = 1/16$, the A.P.s

$$1/16, 2/16, 3/16, 4/16, 5/16, 6/16, \dots$$

Hence, the corresponding H.P is $16, 8, 16/3, 4, 16/5, \dots$

D) Fibonacci sequence

The following sequence of numbers is called Fibonacci sequence

$$0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, \dots$$

which is recursively defined as $F_0 = 0, F_1 = 1$,

and

$$F_n = F_{n-1} + F_{n-2}, \text{ for } n \geq 2. \quad \dots (13.2)$$

The terms of Fibonacci sequence are generally denoted by F_n .

By applying the definition (13.2) above recursively, we get the earlier mentioned sequence

$$0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, \dots$$

Of course, the n^{th} term (say for $n=100$) may be found recursively using (I) above, the following formula, *itself quite a cumbersome one*, can be also used to calculate the n^{th} term directly

$$F_n = \left[\left\{ \frac{1 + \sqrt{5}}{2} \right\}^n - \left\{ \frac{1 - \sqrt{5}}{2} \right\}^n \right] / \sqrt{5}.$$

Sum of first n terms of Fibonacci sequence

$$\sum_{i=1}^n F_i = F_{n+2} - 1. \text{ For all } n.$$

Example 13.11

For the Fibonacci sequence $F_0 = 0, F_1 = 1$, and $F_n = F_{n-1} + F_{n-2}$, for $n \geq 2$.

- Write first ten terms of the Fibonacci sequence
- Find sum of first six terms of the Fibonacci sequence

Solution:

- First ten terms are, 0, 1, 1, 2, 3, 5, 8, 13, 21, 34.
- Using the formula $\sum_{i=1}^n F_n = F_{n+2} - 1$, and taking $n=6$, we get

$$\sum_{i=1}^6 F_n = F_8 - 1 = 13 - 1 = 12.$$

This section will throw some light on various properties of sequences.

13.3.1 Bounded Sequence

- A sequence (S_n) is said to be **bounded above** if there is a real, say C , such that $S_n \leq C \forall n \in \mathbb{N}$
- A sequence (S_n) is said to be **bounded below** if there is a real, say D such that $D \leq S_n \forall n \in \mathbb{N}$
- A sequence (S_n) is said to be **bounded, if it bounded below and also bounded above**. In other words, a sequence (S_n) is said to be bounded, if there are real numbers, say C and D , such that $D \leq S_n \leq C \forall n \in \mathbb{N}$

Example 13.12

- The sequence $(\dots, -10, -8, -6, \dots, 0, 2, 4, 6)$ is bounded above, but not bounded below.
- The sequence $(-10, -8, -6, \dots, 0, 2, 4, 6, \dots)$ is bounded below, but not bounded above.
- The sequence $(-10, -8, -6, \dots, 0, 2, 4, 6)$ is bounded both above and below.

Some interesting results (Proofs in Exercises)

- A) An A.P $(a + (n - 1) d) = (a, a + d, a + 2d, \dots)$, which is non-trivial (i. e., $d \neq 0$),
- is *not* bounded above, if $d > 0$, and
 - is *not* bounded below if $d < 0$.
- B) G. P. $(a r^{(n-1)}) = (a, a r, a r^2, \dots)$, which is non-trivial (i. e., $a \neq 0, 1 \neq r \neq 0$)
- It is *not* bounded above, if $(r > 1, \text{ and } a > 0)$,
 - It is *not* bounded below if $r > 1, \text{ and } a < 0$
 - Bounded neither above, nor below if $r < (-1)$
- C) Fibonacci sequence $(F_n) = (0, 1, 1, 2, 3, 5, 8, \dots)$ is not bounded above, but is bounded below by 0.

13.3.2 Monotone Sequence

- A sequence $(S_n), n \in \mathbb{N}$ is called *monotonically increasing* if $S_n \leq S_{n+1}, \forall n \in \mathbb{N}$. Further, (S_n) is *strictly monotonically increasing sequence*, if $S_n < S_{n+1}, \forall n \in \mathbb{N}$
- A sequence $(S_n), n \in \mathbb{N}$ is called *monotonically decreasing* if $S_{n+1} \leq S_n, \forall n \in \mathbb{N}$. Further, (S_n) is *strictly monotonically decreasing sequence*, if $S_{n+1} < S_n, \forall n \in \mathbb{N}$
- A sequence $(S_n), n \in \mathbb{N}$ is called *monotonic sequence*, if it is *either* monotonically increasing *or* monotonically decreasing.

- Alternating sequence: A sequence (S_n) , $n \in \mathbb{N}$ is called alternating sequence if the successive terms change sign from plus to minus to plus, e.g. $(1, -3, 5, -7, \dots)$, or $(-1, 3, -5, 7, \dots)$.

Example 13.13

- $(1, 3, 5, 7, \dots)$ is a strictly monotonically increasing sequence,
- $(-1, -3, -5, -7, \dots)$ is a strictly monotonically decreasing sequence
- $(1, 1/3, 1/5, 1/7, \dots)$ is a strictly monotonically decreasing sequence
- $(-1, -1/3, -1/5, -1/7, \dots)$ is a strictly monotonically increasing sequence
- $(-1, 1, -3, 3, -5, 5, \dots)$ is neither monotonically increasing nor monotonically decreasing.

Remarks: Bounded sequence vs. Monotonic sequence

- A bounded sequence *may not* be monotonic, e. g. $(1, 2, 1, 2, \dots)$, conversely,
- a monotonic sequence *may not* be bounded, e. g. $(1, 2, 3, \dots)$. *However*,
- there are monotonic sequences which are bounded also, e. g. the sequence $(1 + (1/n))$, $-n \in \mathbb{N}$, is bounded above by 2 and below by 1.

Check Your Progress 2

- For the Arithmetic Progression $7, 4.5, 2, -0.5, -3, \dots$
 - find the 25th term
 - Find the sum of all the terms from 15th to 25th terms, both 15th and 25th terms included.
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.....
- For the G.P sequence $6, 2, 2/3, 2/9, 2/27, \dots$, find sum of first 10 terms
 - For the G.P $18, -6, 2, -(2/3), \dots$
 - Find 15th term of the G.P.
 - Find sum of all the terms between 9th and 15th, including both 9th and 15th.
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.....
- For the A.P with initial term = $1/10$ and common-difference = $-(1/15)$, find the corresponding Harmonic Progression (H.P).
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4. For the Fibonacci sequence $F_0 = 0$, $F_1 = 1$, and $F_n = F_{n-1} + F_{n-2}$, for $n \geq 2$.
- Write first 12 terms of the Fibonacci sequence
 - Find sum of terms from sixth to tenth terms, both inclusive, of the Fibonacci sequence, using appropriate formula for sum of terms.

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5. Give an example of each of the following:

- A strictly monotonically increasing sequence
- A strictly monotonically decreasing sequence
- A sequence which is neither monotonically increasing nor monotonically decreasing.

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6. Give an example of each of the following

- A bounded sequence *is not* monotonic
- A monotonically increasing sequence *is not* bounded
- A sequence which is both monotonic and bounded.

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13.3.3 Convergent, Divergent, Oscillatory Sequences

The *essential idea of convergence of a sequence* (or, that of convergent sequence) is that some sequences (S_n) are well-behaved in the sense that, *except for some initial terms*, all the latter terms of the sequence get *closer and closer* to another, and also to some (fixed) real number, say l . Then, we say the *sequence* (S_n) *converges*, and its limit is l , or alternately we say the *sequence converges to* l .

Formalizing some of the informal notions in the explanation above:

- the notion of '*terms S_n get closer and closer (as we wish) to a number l* ' in mathematics, specially, in analysis/ calculus, is captured by saying '*let $\varepsilon > 0$ be as small as we like, so that distance between the terms S_n and the number l becomes less than or equal to ε , i. e. $|S_n - l| < \varepsilon$* ' Further,

- ii) the notion of ‘*except for some initial terms*’ is captured in mathematics by saying ‘*for all $n \geq n_\varepsilon$* ’, where n_ε is used to indicate that we may leave aside n_ε many terms from being close enough to the limit. The suffix ε in n_ε shows that the number of initial terms to be left aside depends on the choice of ε , because if we want our terms still closer to the limit l , then we have to leave aside more terms: smaller the ε , larger is n_ε : the closer we want to be to the limit (by taking smaller ε), the more is the number of initial terms that may be needed to be left out, and hence, larger the n_ε has to be.

A) Convergent Sequence

A sequence (S_n) is said to be convergent if $\exists a$ (finite) real number l , such that for given $\varepsilon > 0$, $\exists$ a natural number n_ε such that $|S_n - l| < \varepsilon$, for all $n > n_\varepsilon$. Then l is called the limit of the sequence (S_n) , and we write $\lim_{n \rightarrow \infty} S_n = l$.

Example 13.14: Show that the sequence $(S_n) = (1 + 1/n)$ is convergent and converges to 1.

Proof: By definition, the sequence converges to 1 if and only if for given $\varepsilon > 0$, $\exists$ a natural number n_ε such that $|S_n - 1| = |(1 + 1/n) - 1| = |1/n| = 1/n < \varepsilon$, for all $n > n_\varepsilon$ (A)

But by Archimedean property, given positive (real) numbers 1 and ε , $\exists$ a natural number M , such that $M \varepsilon > 1$. And, for all $n > M$, we have $n \times \varepsilon > M \times \varepsilon$. Thus, $n \times \varepsilon > 1$, for all $n > M$. Take $n_\varepsilon = M$, and result is proved.

Some interesting results about convergence and limit

For a given sequence (S_n) , if its limit exists, then it is unique. In other words, if $\lim_{n \rightarrow \infty} S_n = l$, and $l' \neq l$, then l' cannot be a limit of sequence (S_n) .

Proof: Let $|l - l'| = 2\varepsilon > 0$. By the fact that $\lim_{n \rightarrow \infty} S_n = l$, for given $\varepsilon > 0$, $|S_n - l| < \varepsilon$ for all $n > n_\varepsilon$.

Therefore, by triangular law, we have

$$|S_n - l'| = |(S_n - l) + (l - l')| \geq |S_n - l| + |l - l'| \geq 0 + 2\varepsilon > \varepsilon \text{ for all } n.$$

$$|S_n - l'| > \varepsilon \text{ for all } n.$$

Therefore, no n_ε can exist for which $|S_n - l'| < \varepsilon$ for all $n > n_\varepsilon$. Hence, l' cannot be a limit of (S_n) .

According to the above definition, in order to show a sequence to be convergent, if it is actually so, then we somehow have to guess correctly the limit of the sequence. Next, we define the concept of *Cauchy Sequence*, which is equivalent to the concept of convergent sequence and which does not require guessing of the limit. The definition is almost on the similar lines of the definition of convergent sequence without involving limit.

B) Cauchy Sequence

The *essential idea of Cauchy sequence* is the same as that of convergent sequence, i. e., *except for some initial terms*, all the latter terms of the sequence get *closer and closer* to another, rather, as close as we like.

It is already explained how the informal notions involved in the definition of Cauchy's sequence are formalized.

Definition of Cauchy Sequence: A sequence (S_n) is said to be a Cauchy's sequence if for given $\varepsilon > 0$, $\exists$ a natural number n_ε such that $|S_n - S_m| < \varepsilon$, for all $n, m > n_\varepsilon$.

The two concepts of *convergent sequence* and *Cauchy sequence* are equivalent, as shown by the following theorem.

Theorem: A sequence (S_n) is convergent if and only if it is a Cauchy sequence.

C) Non-convergent Sequences

Different categories of *non-convergent* sequences include

i) Finitely oscillatory, (ii) Divergent to $+\infty$, (ii) Divergent to $-\infty$ (iv) Infinitely Oscillatory

ii) *Finitely oscillatory:* A sequence (S_n) is said to be finitely oscillatory, if (a) it is bounded and (b) it is *not* convergent. For example:

$$(S_n) = ((-1)^n) = (-1, 1, -1, 1, \dots); \text{ and}$$

$$(S_n) = (1 + (-1)^n) = (0, 2, 0, 2, \dots)$$

iii) Divergent to $+\infty$: A sequence (S_n) is said to diverge to $+\infty$ if for each real number Δ (*spoken as delta*), howsoever large and positive, there is an n_Δ (number depending on Δ) such that $S_n > \Delta$, for all $n > n_\Delta$. In other words, a sequence (S_n) is *divergent to $+\infty$* if, after some initial terms, its successive terms become larger and larger *unboundedly*.

Example 13.15

a) $(S_n) = ((2n - 1)) = (1, 3, 5, \dots)$

b) $(S_n) = ((a^n))$: $r > 1$, $a > 0$, a G. P. with geometric ratio > 1 , and $a > 0$.

iv) Divergent to $-\infty$: A sequence (S_n) is said to diverge to $-\infty$ if for each real number Δ , howsoever large and positive, there is an n_Δ (number depending on Δ) such that $S_n < (-\Delta)$, for all $n > n_\Delta$. In other words, a sequence (S_n) is *divergent $-\infty$* if its successive members become larger and larger *unboundedly in magnitude* and are *negative in sign*.

Example 13.16

Each of the following sequences diverges to $-\infty$

a) $(S_n) = ((1 - 2n)) = (-1, -3, -5, \dots)$

b) $(S_n) = ((a r^n): r > 1, a < 0)$, a G. P. with geometric ratio > 1 , and initial term $a < 0$.

iv) *Infinitely Oscillatory*: A sequence (S_n) is said to oscillate infinitely, if it is unbounded and diverges neither to $+\infty$ nor to $-\infty$.

Example 13.17

Each of the following sequences is infinitely oscillatory

a) $(S_n) = ((1 + (-1)^{2n})) = (-1, 5, -5, 9, -9, \dots)$

b) $(S_n) = ((a (-r)^n): r > 1, a > 0) = (-a r, ar^2, -a r^3, ar^4, \dots)$, then elements of subsequence $(-a r, -a r^3, \dots)$ become larger and larger in magnitude, but negative in sign, and hence, tend toward $-\infty$, and the subsequence $(ar^2, ar^4, \dots)$ become larger and larger in magnitude with positive sign, and hence, tend toward $+\infty$.

D) Limit Point of a sequence

In the case of the following sequences

i) $(S_n) = ((-1)^n) = (-1, 1, -1, 1, \dots)$; and

ii) $(S_n) = (1 + (-1)^n) = (0, 2, 0, 2, \dots)$

We feel that though these are not convergent sequences (and hence, do not have limit), they apparently have some nice properties, viz, parts of each of the sequences tend to be convergent, e.g., in (i) subsequence $(-1, -1, \dots)$ moves/tends towards -1 ; and subsequence $(1, 1, \dots)$ tends to 1 in (ii) subsequence $(2, 2, \dots)$ tends towards 2 ; and the subsequence $(0, 0, \dots)$ tends to 0 . In view of such sequences, which are not convergent, yet have the above-mentioned property, the concept of *Limit-point* of a sequence, distinct from the concept of Limit of a sequence, is defined as follows.

Limit-point of a sequence: For a sequence (S_n) , a real number, say, LP is said to be a limit point of the sequence (S_n) , if some *subsequence* of it converges to LP .

For example, for the sequence $(S_n) = ((-1)^n) = (-1, 1, -1, 1, \dots)$, the real numbers 1 and -1 are its two limit points.

Results: (i) For a convergent sequence (S_n) , its limit is its limit point also; (ii) Limit of a sequence is its limit point also, but a limit point need not be limit of the sequence. A convergent sequence converges to its limit, the sequence is its own subsequence. Hence, as its own subsequence, it converges to its limit. Therefore, the limit is a limit point.

Converse need not be true: The sequence $(S_n) = (-1, 1, -1, 1, \dots)$ has two limit points, but as it is not convergent, it has no limit.

E) Bolzano-Weierstrass theorem for sequences

Every bounded sequence has a limit point.

Corollary: Every bounded sequence has a convergent subsequence.

F) Examples of convergent and non-convergent sequences

1) Show that the sequence (S_n) , with $S_n = [\sqrt{n+1} - \sqrt{n}]$, for $n \in \mathbb{N}$, is convergent

$$\begin{aligned} \text{Solution: } S_n &= [\sqrt{n+1} - \sqrt{n}] = [\sqrt{n+1} - \sqrt{n}] \times [\sqrt{n+1} + \sqrt{n}] / [\sqrt{n+1} + \sqrt{n}] \\ &= [(\sqrt{n+1})^2 - (\sqrt{n})^2] / [\sqrt{n+1} + \sqrt{n}] \dots [\text{using } a^2 - b^2 = (a+b)(a-b)] \\ &= [(n+1) - n] / [\sqrt{n+1} + \sqrt{n}] \\ &= 1 / [\sqrt{n+1} + \sqrt{n}] \\ &< 1 / [\sqrt{n} + \sqrt{n}] \text{ (As denominator decreases, value increases)} \\ &= 1 / (2\sqrt{n}), \text{ which converges to 0 as follows} \end{aligned}$$

Let $\varepsilon > 0$, to find n_ε such that for all $n > n_\varepsilon$, the following is true

$$|1 / (2\sqrt{n}) - 0| < \varepsilon \text{ squaring both sides, we get}$$

$$1 / 4n < \varepsilon^2 \text{ implying } n > 1 / 4\varepsilon^2$$

Thus, for all $n > 1 / 4\varepsilon^2$,

$$|S_n - 0| < |1 / (2\sqrt{n}) - 0| < \varepsilon \text{ for all } n > 1 / 4\varepsilon^2, \text{ implying}$$

$$\lim_{n \rightarrow \infty} S_n = 0.$$

2. Show that $\lim_{n \rightarrow \infty} [(n^2 + 5n - 7) / ((3n^2 + 12n + 14))] = 1/3$

Solution: Consider

$$\begin{aligned} &[(n^2 + 5n - 7) / ((3n^2 + 12n + 14))] - 1/3 \\ &= [\{3(n^2 + 5n - 7) - (3n^2 + 12n + 14)\} / 3((3n^2 + 12n + 14))] \\ &= [(3n^2 - 35) / 3((3n^2 + 12n + 14))] < 3n / [3((3n^2 + 12n + 14))] < 3n / 9n^2 = 1/3n < 1/n \end{aligned}$$

Thus,

$$|[(n^2 + 5n - 7) / ((3n^2 + 12n + 14))] - 1/3| < |1/n - 0| \text{ (let) } < \varepsilon \dots (i)$$

To find n_ε so that equation (i) above is satisfied by all $n > n_\varepsilon$

$$|1/n - 0| < \varepsilon \text{ is true for all } m, \text{ such that } m > n > 1/\varepsilon = n_\varepsilon.$$

Check Your Progress 3

1) Give an example of each of the following. A sequence which is

- (a) Finitely oscillatory, (b) Divergent to $+\infty$, (c) Divergent to $-\infty$ (d) Infinitely Oscillatory

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